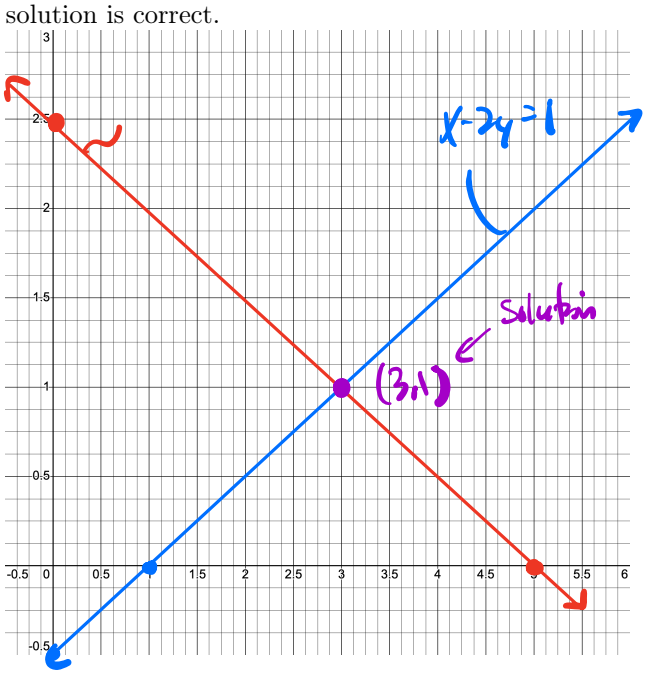
Phase 2 (part 1): Solving Linear Systems of Equations

1.] Consider the following linear system:

$$x - 2y = 1$$
$$x + 2y = 5$$

a.) Solve the linear system for x and y using elimination. Then, graph each line below to confirm the solution is correct.



$$\begin{array}{r} x-2y=1 \\ -(x+2y=5) \\ \hline -4y=-4 \end{array} \rightarrow y=1$$

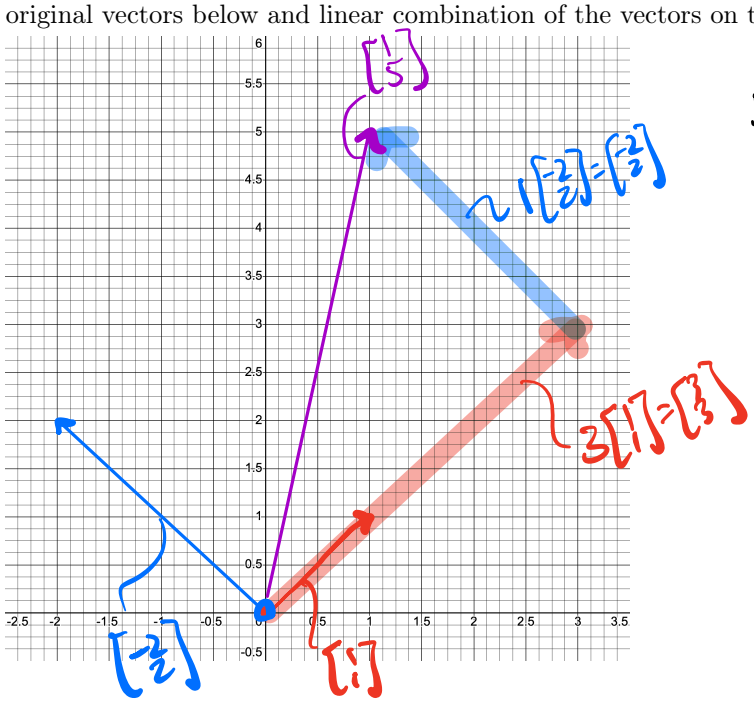
$$x-2(1)=1 \rightarrow x=3$$

Solution: $\boxed{(3,1)}$

$x-2y=1$: x-int: $(1,0)$; y-int: $(0,-1/2)$

$x+2y=5$: x-int: $(5,0)$; y-int: $(0,5/2)$

b.) Write the system as a linear combination, and confirm that the solution above is correct. Sketch the original vectors below and linear combination of the vectors on the plane below:



$$x\begin{bmatrix}1\\1\end{bmatrix}+y\begin{bmatrix}-2\\2\end{bmatrix}=\begin{bmatrix}1\\5\end{bmatrix}$$

$$\underbrace{\begin{bmatrix}1 & -2\\1 & 2\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}x\\y\end{bmatrix}}_{\vec{x}}=\underbrace{\begin{bmatrix}1\\5\end{bmatrix}}_{\vec{b}}$$

2.] Suppose we want to find the equation of the line $y = mx + b$ that goes through the points $(1, -3)$ and $(-4, 5)$. Find the equation of the line by first formulating the problem as a 2×2 linear system of equations. Then, solve it in Python and plot the solution along with the two points to show that the line goes through the points.

$$\begin{cases} -3 = m(1) + b \\ 5 = m(-4) + b \end{cases} \rightarrow \begin{cases} m + b = -3 \\ -4m + b = 5 \end{cases} \rightarrow \left\{ m\begin{bmatrix} 1 \\ -4 \end{bmatrix} + b\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix} \right.$$

$$A\vec{x} = \vec{b}, \quad A = \begin{bmatrix} 1 & 1 \\ -4 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} m \\ b \end{bmatrix}, \vec{b} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

3.]

4.] Suppose we want to find the equation of the plane $z = m_1x + m_2y + b$ that goes through the points $(1, -3, 2)$ and $(-4, 5, 1)$, and $(0, 3, 1)$. Find the equation of the plane by first formulating the problem as a 3×3 linear system of equations. Then, solve it in Python and plot the solution along with the three points to show that the line goes through the points.

$$\begin{cases} 2 = m_1(1) + m_2(-3) + b \\ 1 = m_1(-4) + m_2(5) + b \\ 1 = m_1(0) + m_2(3) + b \end{cases} \rightarrow \begin{cases} m_1 - 3m_2 + b = 2 \\ -4m_1 + 5m_2 + b = 1 \\ 0m_1 + 3m_2 + b = 1 \end{cases}$$

$$\underbrace{\begin{bmatrix} 1 & -3 & 1 \\ -4 & 5 & 1 \\ 0 & 3 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} m_1 \\ m_2 \\ b \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}}_{\vec{b}}$$